

2017 SRJC H2 Chemistry 9729
Preliminary Examination Paper 3 Solution

Section A

- 1 (a)** Oxides of nitrogen, NO_x are a mixture of gases that are composed of nitrogen and oxygen. They are formed during combustion processes in power plants.

Thermal NO_x refers to NO_x formed through high temperature oxidation of the diatomic nitrogen found in combustion air.

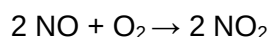
The formation rate is a function of temperature and the residence time of nitrogen at that temperature. At high flame temperatures, molecular nitrogen and oxygen dissociate into their atomic states and participate in a series of reactions.

The two principal reactions producing thermal nitric oxide, NO are as follows.

Step 1 O atoms react reversibly with N₂ to form NO.

Step 2 N atoms liberated react with O₂ to produce more NO.

Nitrogen dioxide typically arises via the oxidation of nitric oxide by oxygen in the atmosphere.



- (i)** Write the two principal reactions involved in the formation of thermal nitric oxide in combustion. [1]



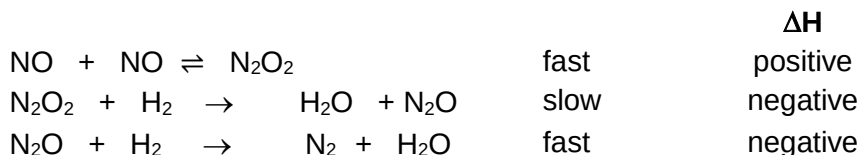
- (ii)** Hence, determine the rate determining step. [1]

Step 1. Step 1 involves the breaking of very strong **N≡N (triple) bond** which requires a large amount of energy to overcome.

- (iii)** Suggest a method to reduce the formation of thermal NO_x. [1]

Reduce peak flame temperature/Reduce time of gas spent in flame.

- (b) Catalytic converters are used to reduce the output of NO_x into the environment. One example is the removal of nitric oxide by reaction with hydrogen gas using platinum catalyst.



- (i) Write the rate equation for the above reaction, showing clearly how you derive it. [2]

$$\text{rate} = k [\text{N}_2\text{O}_2][\text{H}_2]$$

N_2O_2 is an intermediate and should not be included in the rate equation.

Let the equilibrium constant of step 1 be K .

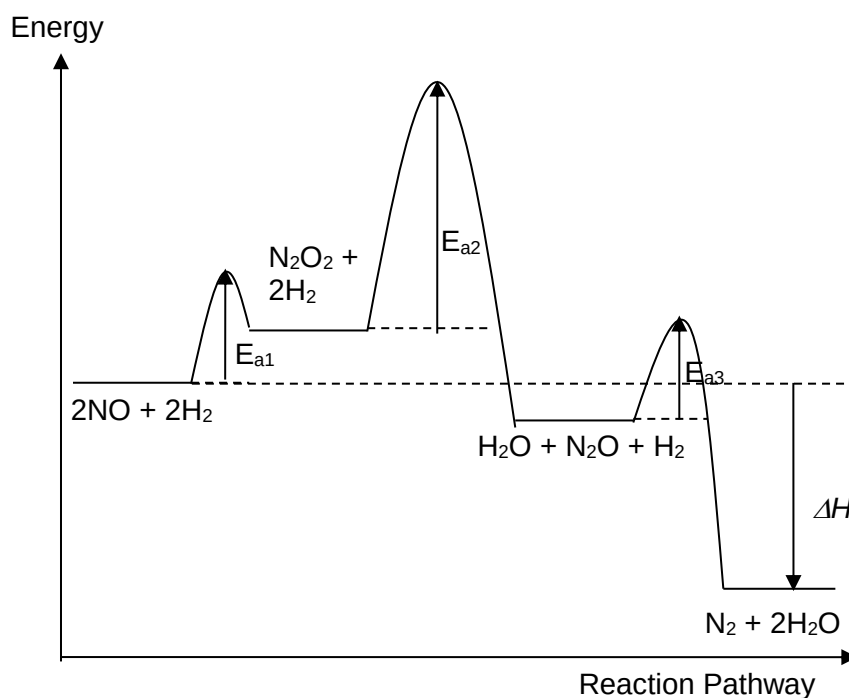
$$K = \frac{[\text{N}_2\text{O}_2]}{[\text{NO}]^2}$$

$$[\text{N}_2\text{O}_2] = K [\text{NO}]^2$$

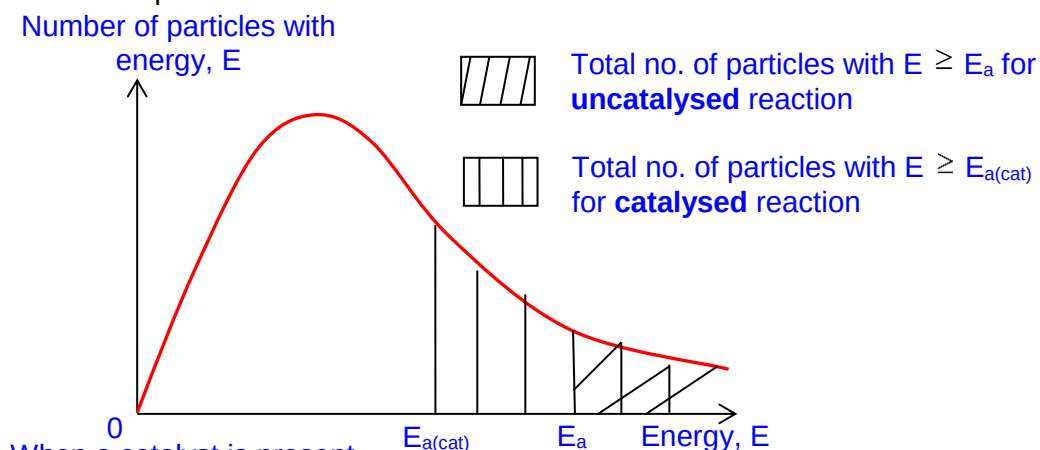
$$\text{rate} = kK [\text{NO}]^2 [\text{H}_2] = k' [\text{NO}]^2 [\text{H}_2]$$

$$\text{where } k' = kK$$

- (ii) Draw the probable energy profile diagram for this overall exothermic reaction. [2]



- (iii) Using an appropriate sketch of the Maxwell–Boltzmann distribution curve, explain how the addition of platinum affects the rate of chemical reaction. [3]



When a catalyst is present,

-activation energy is lowered

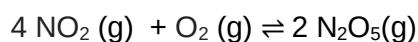
-more reactant particles with energy $\geq E_{a(\text{cat})}$

-more effective collisions

-rate of reaction is proportional to the frequency of effective collisions

rate of reaction increases

- (c) The NO_2 reacts with O_2 in the atmosphere to form colourless nitrogen pentoxide, N_2O_5 in equilibrium.



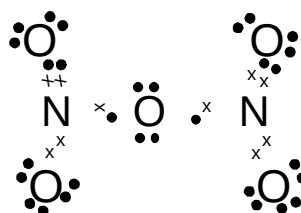
Dinitrogen pentoxide is a rare example of a compound that adopts two structures depending on the conditions. As a gas in the atmosphere, dinitrogen pentoxide exists as a molecule. However, it exists as an ionic salt in the solid state.

- (i) Gaseous N_2O_5 is a symmetrical molecule with the two nitrogen atoms bonded to one central oxygen atom.

Draw a dot-and-cross diagram to show the bonding in N_2O_5 molecule. Use your diagram to:

- State the shape and the bond angle with respect to nitrogen
- Suggest with reasoning, the polarity of the molecule.

[3]



O–N–O bond angle is 120°, trigonal planar

N_2O_5 is polar. The centre O atom is bent. Hence the dipole moment does not cancel out/ there is net dipole moment.

- (ii) A ratio of 4 to 1 mole of gaseous NO_2 and O_2 with a total initial pressure of 2.25 atm was placed in a 1000 cm^3 vessel. The total pressure at equilibrium was 1.5 atm. Calculate K_p for this equilibrium. [3]

Initial pressures

$$P_{\text{NO}_2} = 4/5 \times 2.25 = \underline{1.8 \text{ atm}}$$

$$P_{\text{O}_2} = 1/5 \times 2.25 = \underline{0.45 \text{ atm}}$$

	4 $\text{NO}_2(\text{g})$	+	$\text{O}_2(\text{g})$	$\rightleftharpoons$	2 $\text{N}_2\text{O}_5(\text{g})$
Initial P / atm	1.8		0.45		0
Change P / atm	-2x		-0.5x		+x
Eqm P / atm	1.8 - 2x		0.45 - 0.5x		+x

$$P_{\text{total}} = 1.8 - 2x + 0.45 - 0.5x + x = 1.5 \text{ atm}$$

$$x = 0.500 \text{ atm}$$

$$K_p = \frac{P(\text{N}_2\text{O}_5)^2}{P(\text{O}_2) P(\text{NO}_2)^4}$$

$$K_p = \frac{0.50^2}{0.2 \times 0.8^4}$$

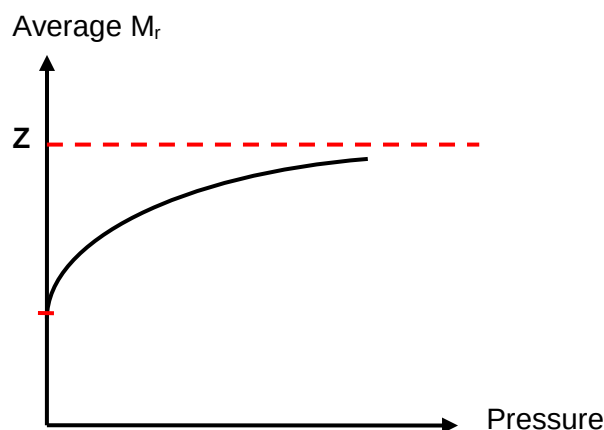
$$K_p = \underline{3.05 \text{ atm}^{-3}}$$

- (iii) Calculate the average M_r of the gases at equilibrium. [1]

Average M_r = Sum of mole fraction $\times M_r$

$$= \frac{0.5}{1.5} \times 108.0 + \frac{0.2}{1.5} \times 32.0 + \frac{0.8}{1.5} \times 46.0 = \underline{64.8}$$

- (iv) The variation of M_r is as shown in the sketch.



Predict a value for Z and account for the shape of the graph. [2]

$$Z = 108.0$$

By LCP, as pressure increases, the position of equilibrium would shift right to decrease the amount of gas. Hence, more N_2O_5 will be produced as average M_r increases.

A curve with decreasing rate and not straight line is obtained as the rate of N_2O_5 formation decreases with decreasing NO_2 and O_2 . Also, the repulsion increases as NO_2 and O_2 molecules are forced closer together with increasing pressure, resulting in a gentler gradient.

The maximum M_r approaches 108.0 but does not reach 108 as the equilibrium mixture must always contain all three gases.

- (d) Solid N_2O_5 dissolved in chloroform may be used to replace concentrated nitric acid and sulfuric acid in the nitration of benzene.

(i) Suggest why solid N_2O_5 can be used in for nitration of benzene. [1]



N_2O_5 contains nitronium ion, NO_2^+ which is an electrophile in the electrophilic substitution of benzene.

- (ii) 0.60 mol of solid N_2O_5 was dissolved in chloroform. Some N_2O_5 decompose back into oxygen and nitrogen dioxide with a rate constant of $1.0 \times 10^{-5} \text{ s}^{-1}$ at 45°C .

The gases produced are collected in a 10 dm^3 container over a period of 40 hours. Calculate the resultant pressure.

[Assume that the products do not dissolve in chloroform.] [3]

Since decomposition follows a first order w.r.t. N_2O_5 (based on units of k)

$$t_{1/2} = \frac{\ln 2}{k} = \frac{\ln 2}{1.0 \times 10^{-5}} = 6.931 \times 10^4 \text{ s}$$

$$\text{Number of } t_{1/2} = \frac{40 \times 60 \times 60}{6.931 \times 10^4} = \underline{2.08}$$



After 40 hours, $0.60 \times 0.75 = \underline{0.45 \text{ mol}}$ of N_2O_5 has decomposed.

$$n_{(\text{gaseous products})} = 2(0.45) + \frac{1}{2}(0.45) = 1.125 \text{ mol}$$

$$pV = nRT$$

$$p = \frac{1.125 \times 8.31 \times 318}{10 \times 10^{-3}} = \underline{2.97 \times 10^5 \text{ Pa}} = 2.97 \text{ bar}$$

Alternative answer

$$n_{\text{N}_2\text{O}_5} = 0.60 \left(\frac{1}{2}\right)^{2.08} = 0.1419$$

After 40 hours, $0.60 - 0.1419 = 0.4581 \text{ mol}$ of N_2O_5 has decomposed.

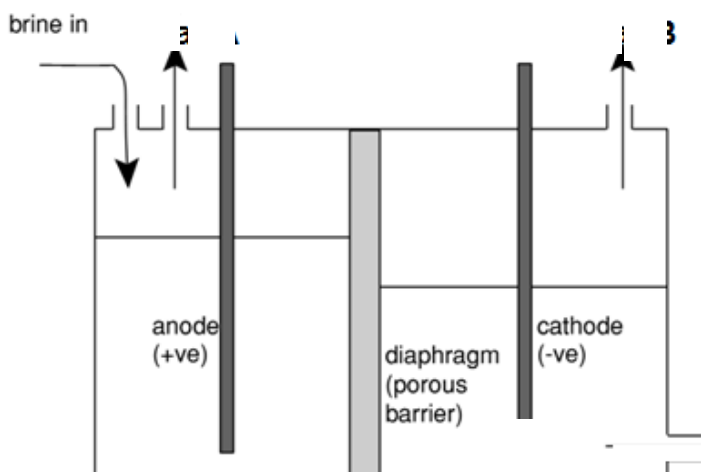
$$n_{(\text{gaseous products})} = 2(0.4581) + \frac{1}{2}(0.4581) = 1.145 \text{ mol}$$

$$p = \frac{1.145 \times 8.31 \times 318}{10 \times 10^{-3}} = \underline{3.03 \times 10^5 \text{ Pa}} = 3.03 \text{ bar}$$

- 2 Chlorine is a yellow-green gas that was first synthesised around 1630. Chlorine forms a variety of oxides such as chlorate(VII), ClO_4^- and chlorate(V), ClO_3^- .

- (a) Chlorine gas can be produced from the electrolysis of brine (concentrated sodium chloride solution) using a diaphragm cell with inert electrodes.

A diaphragm is used to keep the products of electrolysis from mixing and hence, prevent the formation of undesirable by-products.



- (i) Write the ion-electron half-equations, with state symbols, for the reactions taking place at the electrodes. [2]

Cl^- (anion) would migrate to the anode and be oxidised to Cl_2 gas.

Anode reaction: $2\text{Cl}^- (\text{aq}) \rightarrow \text{Cl}_2 (\text{g}) + 2\text{e}^-$

Cathode reaction: $2\text{H}_2\text{O} (\text{l}) + 2\text{e}^- \rightarrow \text{H}_2 (\text{g}) + 2\text{OH}^- (\text{aq})$

- (ii) Calculate the mass of chlorine gas produced if a current of 5 A was passed for 7 hours through the electrolytic cell. [2]

$$Q = I \times t = 5 \times 7 \times 60 \times 60 = 1.26 \times 10^5 \text{ C}$$

$$Q = n_e \times F$$

$$\text{Amount of electrons passed} = \frac{1.26 \times 10^5}{96500} = 1.306 \text{ mol}$$

$$\text{Amount of chlorine} = \frac{1.306}{2} = 0.653 \text{ mol}$$

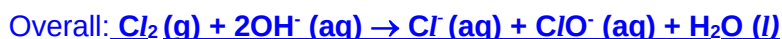
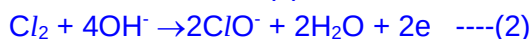
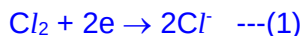
$$\text{Mass of chlorine produced} = 0.653 \times 2 \times 35.5 = \underline{\underline{46.4 \text{ g}}}$$

- (b) In some industrial electrolytic cells, the products of electrolysis of brine are allowed to react further when the diaphragm is removed. Two chlorine-containing products are formed, depending on the operating conditions used. At low temperatures, one of the products is a chloro-oxoanion with an oxidation state of +1.

(i) State the type of reaction taking place. [1]

Disproportionation

(ii) Give an overall balanced equation for the reaction described above. [1]



- (c) With reference from the Data Booklet, explain why dilute aqueous sodium chloride cannot be used as the electrolyte for the manufacture of chlorine in this electrolysis. [1]

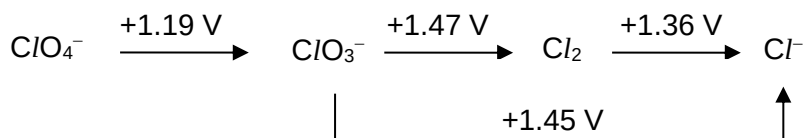


When dilute sodium chloride is used, the $E^\ominus(\text{O}_2/\text{H}_2\text{O})$ is less positive than $E^\ominus(\text{Cl}_2/\text{Cl}^-)$ and H_2O is preferentially oxidised at the anode to give O_2 gas, instead of chlorine gas.

- (d) The standard electrode potentials, $E^\ominus$, and standard Gibbs free energy changes, $\Delta G^\ominus$, of different chlorine-containing species are tabulated below.

	Half-equation	$E^\ominus / \text{V}$	$\Delta G^\ominus / \text{kJ mol}^{-1}$
1	$\text{ClO}_4^- + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{ClO}_3^- + \text{H}_2\text{O}$	+1.19	-230
2	$2\text{ClO}_3^- + 12\text{H}^+ + 10\text{e}^- \rightleftharpoons \text{Cl}_2 + 6\text{H}_2\text{O}$	+1.47	?
3	$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	+1.36	-262

These electrode potentials can be summarised using the Latimer diagram shown below. In a Latimer diagram, the most highly oxidised form of the element is on the left, with successively lower oxidation states to the right. The electrode potentials are shown on the arrows.



(i) Calculate $\Delta G^\ominus$ for half-equation 2. [1]



$$\Delta G^\ominus = -nFE^\ominus$$

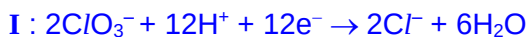
$$= -(10)(1.47)(96500) = -1418.6 = \underline{-1420 \text{ kJ mol}^{-1}}$$

- (ii) The standard electrode potential of converting ClO_3^- to Cl^- is found to be +1.45 V, instead of +2.83 V. The number of electrons transferred in each step must be taken into account.

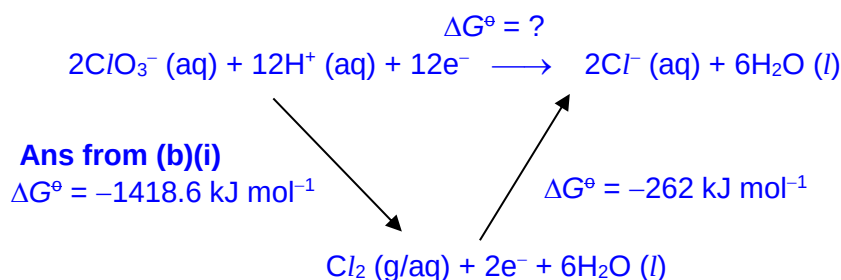
I: Write a half-equation for the conversion of ClO_3^- to Cl^-

II: Using your knowledge of Hess' Law for ΔG° and your answer to (d)(i), show with the aid of an energy cycle that the E° for the conversion of ClO_3^- to Cl^- is +1.45 V.

[4]



Or

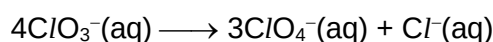


Applying Hess Law,

$$\Delta G^\circ = -1418.6 + (-262) = \underline{-1680.6 \text{ kJ mol}^{-1}}$$

$$E^\circ = -[-1680.6 \times 10^3 / (12 \times 96500)] = \underline{+1.45 \text{ V}} \text{ (shown)}$$

- (iii) With the help of the Latimer diagram provided, calculate E°_{cell} of the reaction below. Hence, determine its $\Delta G^\circ_{\text{cell}}$.



[2]

Using Latimer diagram: $E^\circ_{\text{cell}} = +1.45 - (+1.19) = \underline{+0.26 \text{ V}}$

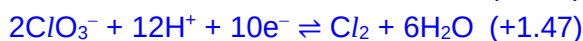
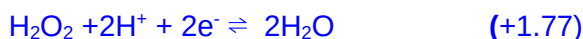
1 ClO_3^- ion is reduced to Cl^- , gaining 6 electrons.

3 ClO_3^- ions are oxidised to 3 ClO_4^- ions, losing $3 \times 2 = 6$ electrons

Amount of electrons transferred = 6 mol

$$\Delta G^\circ_{\text{cell}} = -(6)(96500)(+0.26) = \underline{-151 \text{ kJ mol}^{-1}}$$

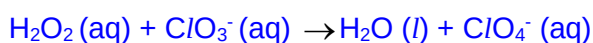
- (iv) With reference to the *Data Booklet*, predict what will happen when an excess of acidified hydrogen peroxide solution is added to a solution of Cl^- (aq). [3]



$$E^\circ_{\text{cell}} = (+1.77) - (1.36) = \underline{+0.41 \text{ V}} > 0 \therefore \text{reaction is feasible.}$$



$$E^\circ_{\text{cell}} = (+1.77) - (+1.47) = \underline{+0.30 \text{ V}} > 0 \therefore \text{reaction is feasible.}$$



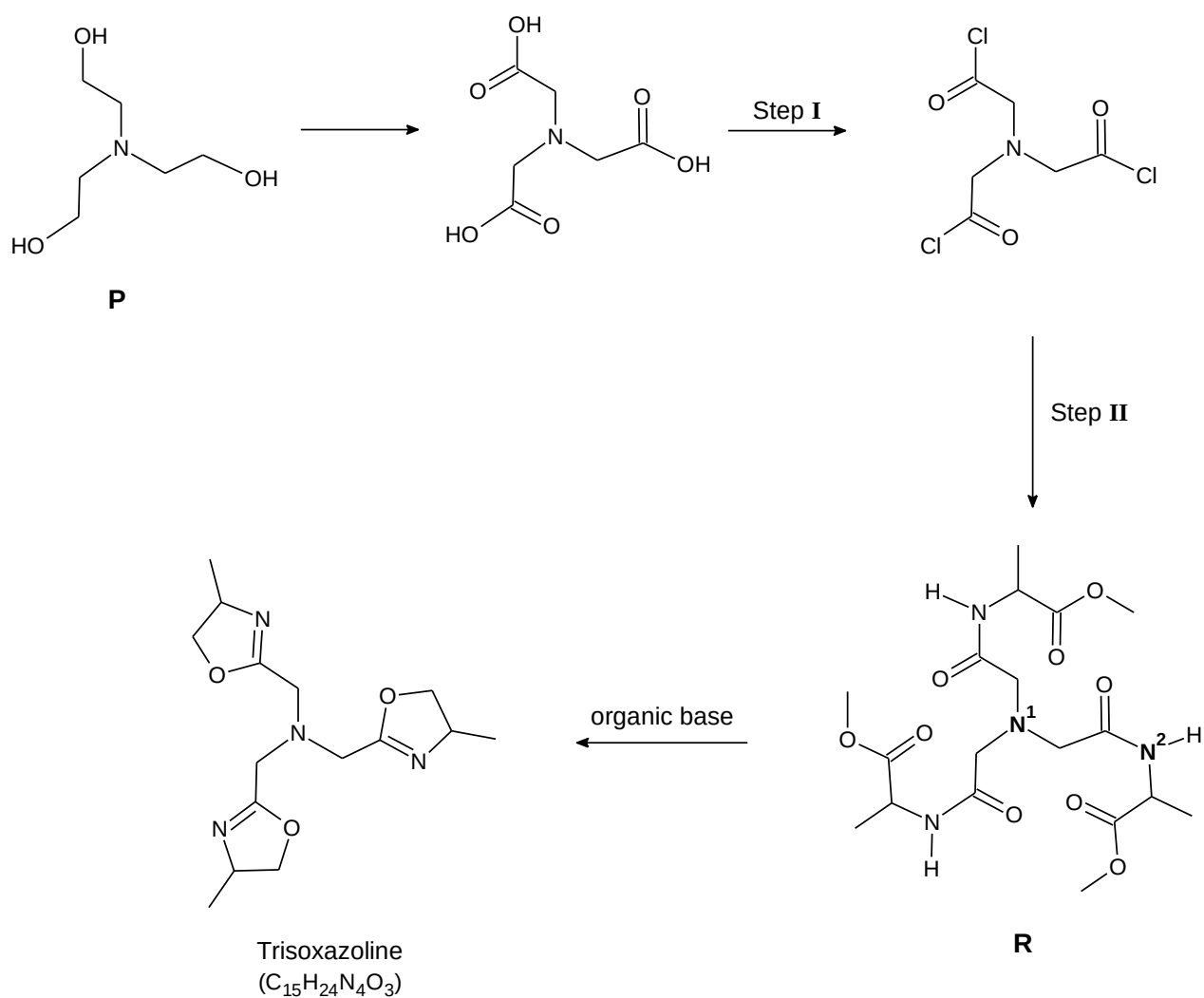
$$E^\circ_{\text{cell}} = (+1.77) - (+1.19) = \underline{+0.58 \text{ V}} > 0 \therefore \text{reaction is feasible.}$$

Cl^- will be oxidised to Cl_2 , then to ClO_3^- and finally to ClO_4^- while H_2O_2 will be reduced to H_2O .

[Total: 17]

- 3 Trisoxazoline are organic molecules that can function as ligands. Despite their huge molecular structure, they are able to form stable complexes with metals. Metal complexes with trisoxazoline are commonly used as catalyst in organic synthesis.

(a) The reaction scheme below illustrates the synthesis of trisoxazoline.



- (i) Suggest the reagents and conditions required for step I and step II. [2]

Step I: $\text{PCl}_5(\text{s})$, rtp or $\text{PCl}_3(\text{s})$, rtp or $\text{SOCl}_2(\text{l})$, rtp

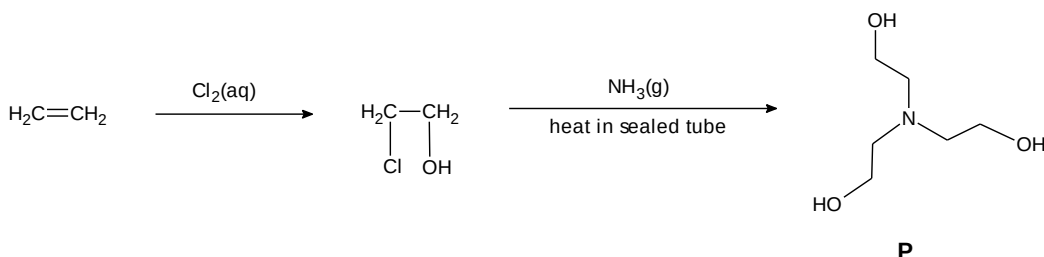
Step II: CC(C)C(=O)N, rtp

- (ii) Explain the difference in basicity for N¹ and N² present in **R**. [2]

N² on amide is neutral as the lone pair of electrons is delocalised with the π-electrons of C=O. Hence lone pair of electrons not available to form dative bond with proton.

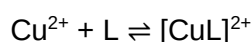
N¹ atom is basic as the lone pair of electrons is available to form dative bond with proton.

- (iii) Propose a 2-step reaction scheme to synthesise compound **P** from ethene. [2]



Trisoxazolines are usually applied to transition metals, such as copper, to form complexes. These complexes are useful in advanced organic synthesis to make specific isomers.

Table 1 shows some properties of the copper complexes, where K_{stab} refers to the stability constant of the complex, e.g.



$$K_{\text{stab}} = \frac{[\text{CuL}]^{2+}}{[\text{Cu}^{2+}][\text{L}]}$$

Table 1

Complex	log(K_{stab})	Colour	Shape of complex
copper(II)-trioxazoline	12.9	green	distorted tetrahedral
copper(II)-EDTA	18.8	blue	octahedral
copper(II)-en	18.7	violet	octahedral

- (b) Describe the feature of the trioxazoline molecule that enable it to act as polydentate ligand. Hence, state the type of bond formed between the ligand and the central Cu(II) ion and give the coordination number of the complex. [2]

The presence of 4 N atoms with lone pairs available to form dative bonds with metal centre. Coordination number is 4

- (c) (i) Using copper(II)-EDTA as an example, explain why the five d-orbitals are split into two different energy levels. [2]

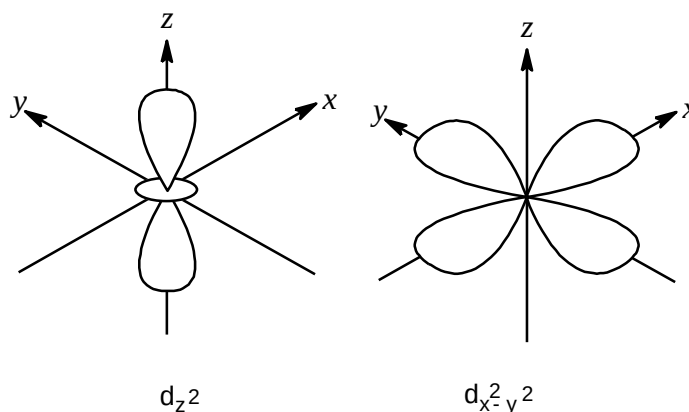
The lone pair of electrons on EDTA approaches the Cu²⁺ along the axis of octahedral complex.

The d_{x²-y²} and d_{z²} orbitals lie along the x, y, z axis will experience greater inter-electronic repulsion as the electrons in these d orbitals are pointing towards the lone pairs of electrons from EDTA.

Hence, the five d orbitals will be split into 2 energy levels - the d_{xy}, d_{yz} and d_{xz} orbitals at a lower energy level and the d_{z²} and d_{x²-y²} orbitals at a higher energy level.

(ii) Draw the shapes of the d-orbitals at the higher energy level.

[1]



(iii) Explain why the copper(II) complexes are coloured.

[3]

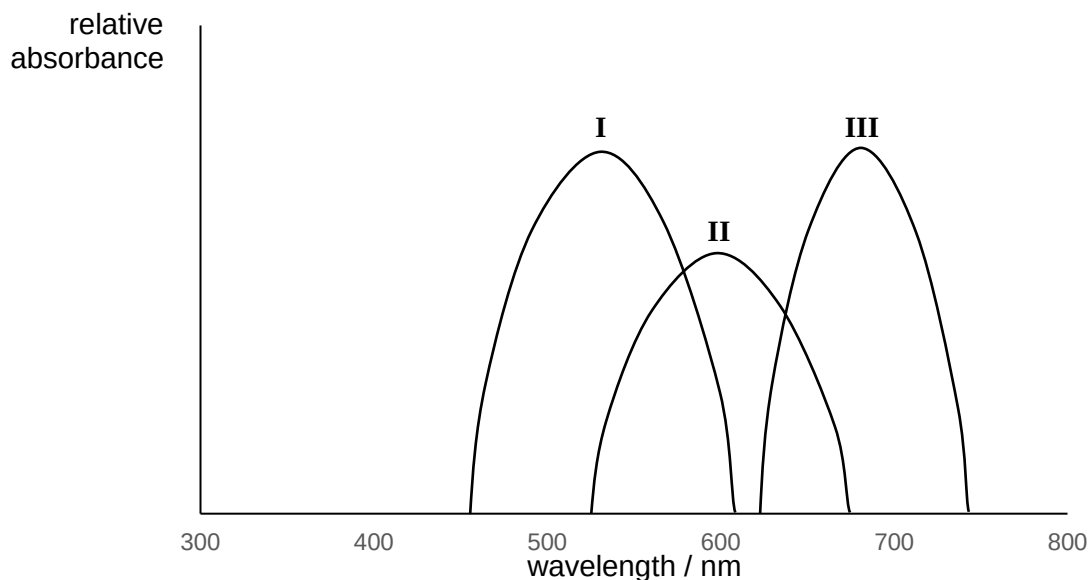
Presence of ligands splits the d-orbitals into 2 energy levels.

1 of the d-electrons absorbs certain wavelength from the visible light spectrum and undergoes d-d transition.

The d-electron is promoted to a higher energy level. The remaining wavelengths are transmitted and the complementary colour of the wavelength absorbed is observed.

(iv) The visible absorption spectrum of the three complexes is shown below where each peak represents one complex.

[A visible absorption spectrum is a graph depicting the absorption of radiation by a material over a range of visible light wavelengths.]



Identify the copper(II) complex responsible for each absorption peak. Hence, arrange the three complexes in order of increasing magnitude of their energy gap.

[2]

I: copper(II)-en

II : copper(II)-EDTA

III : copper(II)-trioxazoline

Since $\Delta E \propto \frac{1}{\lambda}$, energy gap in increasing order:

copper(II)-trioxazoline < copper(II)-EDTA < copper(II)-en

- (d) Using Table 1, comment on the spontaneity of forming the respective copper(II) complexes. [2]

$$\Delta G^\circ = -RT \ln K$$

K_{stab} : copper(II)-EDTA > copper(II)-ethylenediamine > copper(II)-trioxazoline

Magnitude of ΔG° :

copper(II)-EDTA > copper(II)-ethylenediamine > copper(II)-trioxazoline

spontaneity: **Cu(II)-EDTA > Cu(II)-en > Cu(II)-trioxazoline**

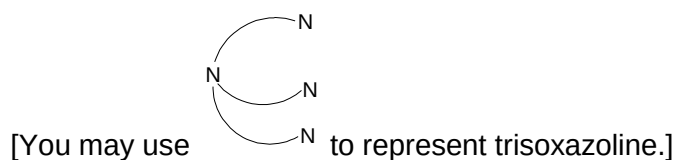
- (e) Other than copper(II) ions, it was found that copper(I) ions are also able to form complexes with trioxazoline.

- (i) Predict and explain the colour of solid copper(I)-trioxazoline complex. [1]

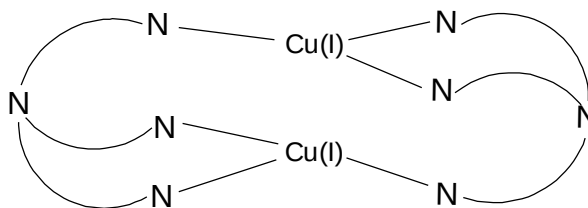
White. The **d-orbitals are fully filled** in copper(I), **no d-d transition**.

- (ii) Analysis was done on copper(I)-trioxazoline complex. The relative molecular mass of the complex was apparently 743.0. Further analysis showed that each copper(I) ion is bonded to three nitrogen atoms.

Propose the structure of copper(I)-trioxazoline complex.



[1]



[Total: 20]

Section B

- 4 (a) At r.t.p, 1.2 dm³ of chlorine gas is reacted with 6.7 g of compound **A**, C₉H₁₀O, to give compound **B**. On addition of PCl₅, **A** gives white fumes. Treatment of **A** with NaBr and concentrated H₂SO₄ under reflux gives a mixture of compounds **C** and **D**. **C** and **D** have the same molecular formula C₉H₁₀Br₂ and each contains only *one* chiral centre.

- (i) When NaBr is mixed with concentrated H₂SO₄, the following observations are made.

- White fumes produced
- Reddish brown liquid formed
- Pungent gas decolourised purple KMnO₄

It is suggested that a displacement reaction occurred first, producing white fumes. Some of the white fumes further react with H₂SO₄ via a redox reaction. Using the above information, write the two corresponding chemical equations for this reaction. [2]



- (ii) Using your answer in (a)(i), deduce the identities of **A** to **D**. [5]

Amt of **A** = 6.7/134 = 0.05 mol

Amt of Cl₂ = 1.2/24 = 0.05 mol

A has similar ratio of C and H atoms, and no. of C > 6

→ **A** contains a benzene ring

A undergoes electrophilic addition with chlorine gas

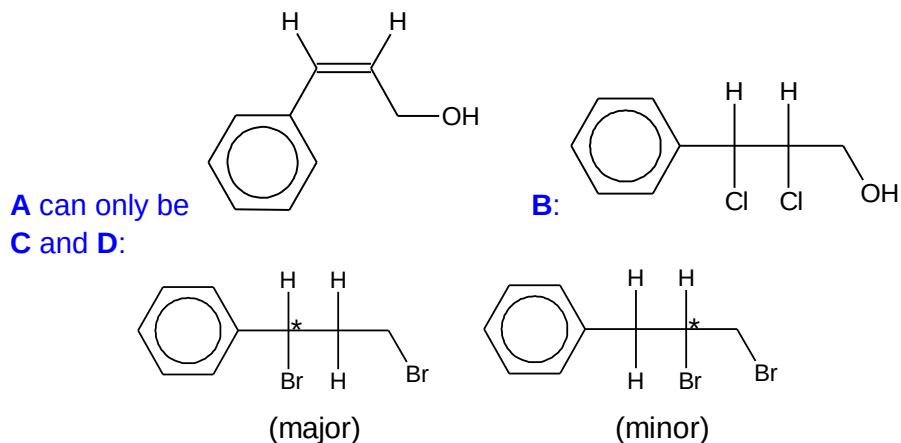
→ **A** contains 1 alkene functional group (or C=C) and **B** is a halogenoalkane/chloroalkane

A undergoes nucleophilic substitution with PCl₅ to give white fumes of HCl

→ **A** is an alcohol

A undergoes both electrophilic addition and nucleophilic substitution to give **C** and **D**.

Since **C** and **D** each only contains 1 chiral centre,



- (iii) Account for the relative quantity of compound **C** and **D** formed.
Compound **C** is the major product while **D** is the minor product.

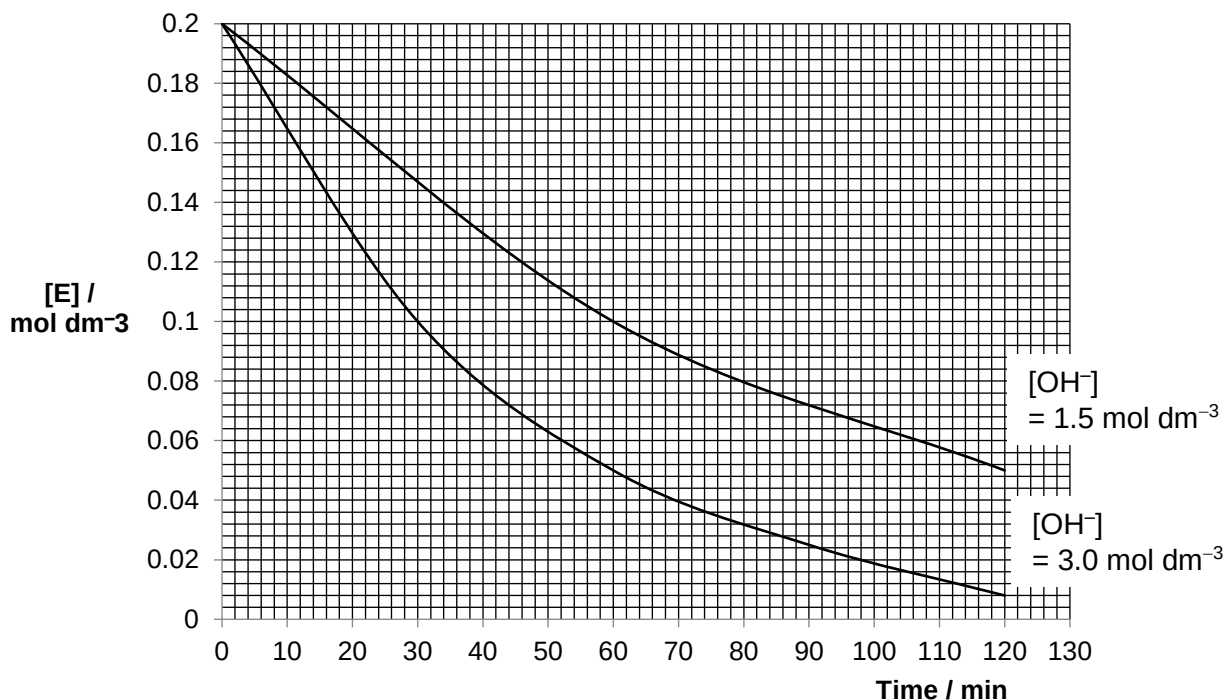
[1]

For compound **D**, the benzylic carbocation intermediate is more stable due to resonance / positive charge dispersed into the benzene ring.

- (b) **E** is another constitutional isomer of **C** and **D**.
A series of chemical experiments were conducted on **E**.

Experiment 1:

The reaction kinetics of **E** with aqueous sodium hydroxide was determined by monitoring the change in concentration of **E** with time.



Experiment 2:

0.2 mol of **D** and **E** were treated separately with boiling aqueous sodium hydroxide. The products from each compound were then acidified with nitric acid and then treated with silver nitrate solution.

The results obtained are shown below:

	Observation upon adding AgNO_3	Mass of ppt / g
D	formation of cream-coloured precipitate	75.12
E	formation of cream-coloured precipitate	37.56

- (i) Using experiment 1, deduce the rate equation.
using $[\text{OH}^-] = 3.0 \text{ mol dm}^{-3}$ graph,
 $t_{1/2}$ is constant at 30 min, reaction is **1st order with respect to E**.
OR
using $[\text{OH}^-] = 1.5 \text{ mol dm}^{-3}$ graph,
 $t_{1/2}$ is constant at 60 min reaction is **1st order with respect to E**.

[3]

Using initial rate method,
Initial rate for graph where $[\text{OH}^-] = 1.5 \text{ mol dm}^{-3}$
 $= 0.2/110 = \underline{0.001818 \text{ mol dm}^{-3} \text{ min}^{-1}}$

Initial rate for graph where $[\text{OH}^-] = 3.0 \text{ mol dm}^{-3}$

$= 0.2/52 = 0.003846 \text{ mol dm}^{-3} \text{ min}^{-1}$

As $[\text{OH}^-]$ increases 2 times, initial rate also increases 2 times

$$\left(\frac{0.004}{0.00211} = 1.90 \approx 2\right)$$

Hence, **order of reaction with respect to OH^- is 1.**

Rate = $k [\text{E}] [\text{OH}^-]$

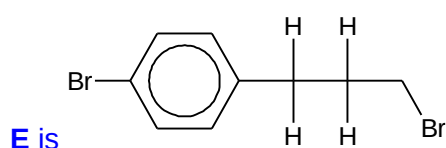
- (ii) Using Experiment 2 and (b)(i), propose a possible structure of **E** and draw the mechanism for the reaction of **E** with NaOH(aq) . [3]

From (b)(ii), **E** undergoes $\text{S}_{\text{N}}2$ mechanism

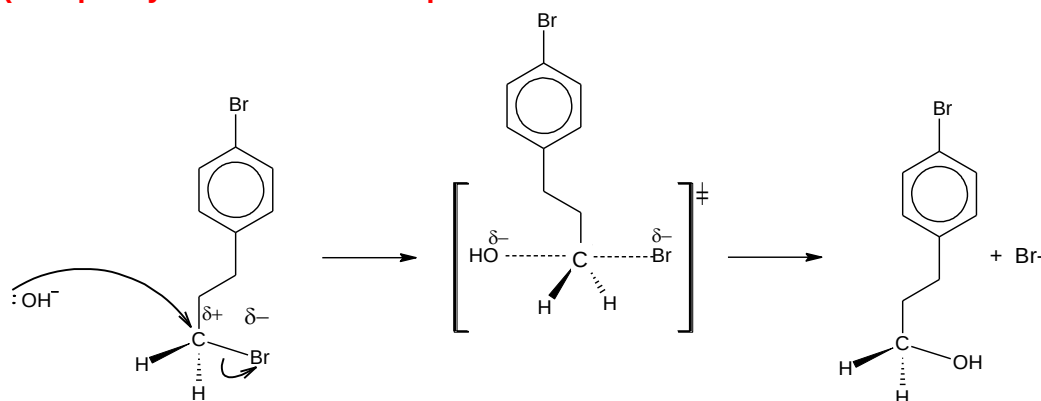
→ it is a **1° halogenoalkane**

From expt 2, 0.2 mol of **E** forms 0.2 mol of AgBr

→ There is **only one** Br in the alkyl side chain



(accept any mono substituted position of Br on benzene with 1° or 2° bromoalkane)



- (c) Depending on the mechanism, the solvent affects the stability of nucleophile and/or reaction intermediates in the nucleophilic substitution of halogenoalkanes. The table below shows the rate of reaction when different halogenoalkanes and solvents are used.

Solvent	Type	Dielectric constant, ϵ	Relative rates for reaction with OH^-	
			1° RX in $\text{S}_{\text{N}}2$	3° RX in $\text{S}_{\text{N}}1$
CH_3OH	Protic	33	1	4
H_2O	Protic	78	7	150 000
CH_3COOH	Protic	6	1	1
CH_3COCH_3	Aprotic	21	5000	-

Dielectric constant, ϵ , is a measure of the solvent polarity and ability to insulate charge.

- (i) Suggest, using **only** structure and bonding, the difference between *protic* and *aprotic* solvents. [1]

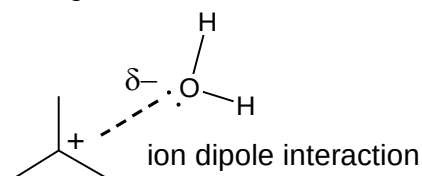
Protic refers to molecules with H atoms that can be used to form hydrogen bond. Aprotic solvents can only form permanent dipole- permanent dipole interactions between molecules/ have no H atoms that can form hydrogen bond

- (ii) Explain the effect of solvent on the relative rate of S_N2 reactions. [2]

For S_N2 , protic solvent like water can form hydrogen bond with OH^- . Hence decreasing the nucleophilic ability of OH^- and lower the rate of reaction.

While aprotic solvent like CH_3COCH_3 , do not form hydrogen bond and hence the lone pair of electron on OH^- is more available to attack the electron deficient C on 1° RX.

- (iii) Explain, with an aid of a diagram, how water increases the rate of S_N1 reaction. [2]



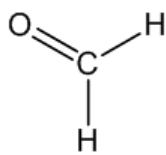
The ion dipole interaction stabilised the carbocation/reaction intermediate

- (iv) Name a solvent, other than those given in the table, that will result in a slower rate for S_N1 than ethanoic acid. [1]

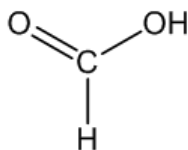
Name of any suitable hydrocarbon e.g Benzene, hexane ..etc

[Total: 20]

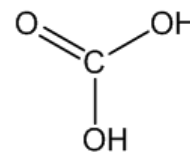
- 5 (a) Using only the elements C, H and O, draw the structural formulae of **three** organic compounds, each containing a single carbon atom with an oxidation state of zero, +2 and +4 respectively. [3]



O.S. = 0



O.S. = +2



O.S. = +4

- (b) Hot, concentrated potassium manganate (VII) oxidises several classes of organic compounds to ketones, carboxylic acids or carbon dioxide. By this means, the structures of compounds can be determined. Some compounds are easily oxidised, while others require longer heating.

The following describes some reactions of compounds **F** and **K**, and of their oxidation products.

F, $C_8H_{12}O$ reacts with excess potassium manganate (VII) to produce single organic products, **G**, $C_4H_6O_5$ while **K**, $C_{12}H_{12}$, reacts with the same reagent to produce **H**, $C_{10}H_{10}O_3$. Carbon dioxide is produced in both reactions in a mole ratio of 2 : 1 respectively. During oxidation of **F**, four moles of carbon dioxide were liberated.

Although **F** reacts with potassium manganate (VII), it gives no reaction with potassium dichromate (VI). When 0.10 mol of **F** is reacted with an excess of sodium metal, 1.2 dm³ of hydrogen is formed, measured at room temperature and pressure.

G reacts with excess concentrated sulfuric acid to give **I**, $C_4H_4O_4$. Upon further oxidation, **I** is found to give **J**, $C_3H_2O_5$, and an inorganic by-product.

H gives a positive iodoform test and dissolves in aqueous sodium hydroxide. Upon further oxidation of **H**, **L**, $C_9H_6O_6$, and a similar inorganic by-product formed from **I** is also produced.

- (i) Deduce the structure of compounds **F** to **L**, explaining the chemistry of the reactions described. [11]

F and **K** undergoes strong oxidation with $KMnO_4$ to give **G** and **H**

→ Since there is a decrease in no of C atoms after oxidation, both **F** and **K** must contain alkenes.

F liberates four mole of carbon dioxide

→ **F** contains (terminal) alkenes.

F undergoes redox with Na metal

→ $0.10\text{mol} \equiv 1.2\text{dm}^3 \equiv 1H_2 \equiv 1ROH$

→ **F** contains only 1 alcohol.

F does not undergo strong oxidation with $K_2Cr_2O_7$

→ **F** contains a tertiary alcohol.

F undergoes elimination with excess concentrated sulfuric acid to give **I**, $C_4H_4O_4$

→ **F** contains an alcohol.

I further oxidises to give J, $C_3H_2O_5$, and an inorganic by-product
 → I contains 4 carboxylic acid and CO_2 is the by-product

H undergoes mild oxidation with iodoform
 → H contains a terminal methyl ketone
 (since it is an [O] product, it cannot be a methyl carbinol)

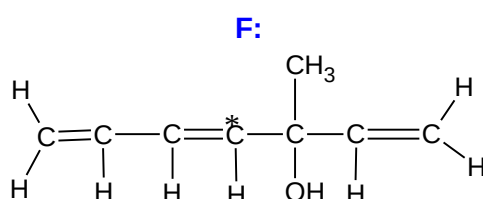
H dissolves in aqueous sodium hydroxide.
 → H contains an acidic group (i.e. $-COOH$)

L is an oxidised product
 → L contains a tricarboxylic acid and the inorganic product is CO_2 .

G	I	J
K	H	L

Note : methyl group can be on any position

- (ii) State the type(s) of stereoisomerism shown by compound F and give one further piece of relevant information about it. [2]



Enantiomerism:

F has no plane of symmetry. OR
 F rotates plane polarised light. OR
 F has a pair of non-superimposable images.

Cis-trans:

Restricted rotation about $C=C$ OR
two different substituents about each C on $C=C$

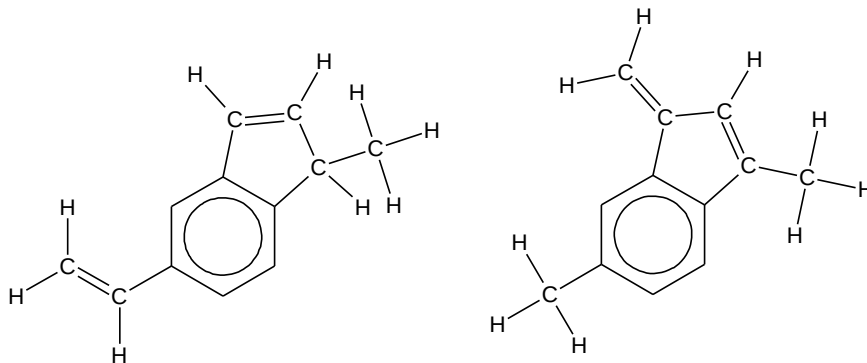
- (iii) Hence, predict the total number of isomers shown by F. [1]

no of enantiomers present = 4

- (iv) When compound **K** undergoes prolonged heating with KMnO_4 to give **L**, **K** exhibits **two** types of constitutional isomerism.

Name the specific types of isomerism shown by compound **K** and hence, draw the **displayed formulae** of the isomers. [3]

Positional and Chain isomerism



[Total : 20]